

DMV302

Data Mining & Visualizations



Component Details	
Component code and name	DMV302 – Data Mining & Visualizations
NZQA Level	Level 7
Credits	15
Workload	150 hours over 16 weeks

Component Description and Aims

This component aims to teach students data mining techniques for both structured and unstructured data. Students will be able to analyse moderate-to-large sized datasets, data preparation, handling missing data, modelling, prediction and classification. Students will also be able to communicate complex information in results of data analytics through effective visualization techniques.

- Learning Outcomes
- On successful completion of this component students are able to:
- a) Appraise the available approaches, techniques and categories for data mining problems.
 - b) Contrast and assess appropriate techniques for detecting and evaluating patterns in a given data set.
 - c) Generate statistical predictions using regression models, neural networks, or tree-based methods.
 - d) Assess key components of the computing environment for data mining, visualization, and association analysis including software, hardware, and analytical tools.
 - e) Evaluate a range of visualization techniques and Select an appropriate technique to properly present data to different audiences.

Workload and Engagement

Students are expected to spend up to 10 hours a week working on this course. Students have 4 hours of contact available to them each week for 16 weeks which consists of attending class, responding to

lecturer feedback, contributing to class discussion, and working on projects. In addition to this, students are expected to work on prescribed readings and assignments for the remainder of their time.

Content Outline

- Structured and unstructured data analysis,
- Datasets of different sizes, data preparation, modeling,
- Classification and modeling, and handling missing data;
- Data analysis for effective visualization and its techniques.

Assignments

The component has three summative assessments as given below:

Assessment	Weight	Week Issued	Week Due
Data Analysis	25.00%	Week 3	Week 7
Cluster Analysis	35.00%	Week 8	Week 12
Anomaly detection and Association analysis	40.00%	Week 12	Week 16

Dates are subject to change with prior notice.

Assignment Overviews

Assignment One: Data cleaning and Data visualization—Report (25%)

In this assessment, students undertake two tasks, firstly a case study and write a report advising a client of any potential data mining applications they can employ in their business. Secondly, discover and explore data from the stock market time series datasets.

Assignment Two: Cluster Analysis (35%)

This assessment requires you to implement a few data mining algorithms for clustering and classification to solve problems in real-world scenarios. You will also need to answer a few questions in this problem set.

Assignment Three: Anomaly detection and Association analysis (40%)

This assessment requires you to individually complete data mining and visualization tasks and creatively and programmatically visualize a data set. Perform association analysis on transactional data.

Grading of Assessments

The below chart outlines the grading scale to be used for grading student assignments and for final grading. This information will appear on a student's academic transcript.

Grade	Range	Values
A+	100-90%	100%
A	89%-85%	88%
A-	84%-80%	83%
B+	79% - 75%	78%
B	74% - 70%	73%
B-	69%-65%	68%
C+	64% - 60%	63%
C	59% - 55%	58%
C-	54% - 50%	53%
D	49% - 0%	0%

Recommended Learning Sources

Books

- Dean, J. (2014). Big Data, Data Mining, and Machine Learning: Value Creation for Business Leaders and Practitioners. John Wiley and Sons.
- Knaflic, C. (2015). Storytelling With Data: A Data Visualization Guide for Business Professionals: John Wiley and Sons
- Marr, B. (2016). Data in Practice: How 45 Successful Companies Used Big Data Analytics to Deliver Extraordinarily. John Wiley and Sons.

Online

- Sharma, R. (2021, 8 January). 12 most useful data mining applications of 2021. <https://www.upgrad.com/blog/12-most-useful-data-mining-applications-of-2020/>
- Altalhi, A. H., Luna, J. M., Vallejo, M. A. & Ventura, S. (2017). Evaluation and comparison of open source software suites for data mining and knowledge discovery. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 7(3), e1204. https://www.researchgate.net/publication/314151385_Evaluation_and_Comparison_of_Open_Source_Software_Suites_for_
- Fayyad, U., Piatetsky-Shapiro, G. & Smyth, P. (1996). From data mining to knowledge discovery in databases. *AI Magazine*, 17(3), 37–37. <https://www.kdnuggets.com/gpspubs/aimag-kdd-overview-1996-Fayyad.pdf>
- Data Science Project Management. What is CRISP DM? <https://www.datascience-pm.com/crisp-dm-2/>

Submission Policy

Please refer to the Submission Policy on MyLearn for further details.

NOTE: All information provided in the document is subject to change